3.21 harmonic voltage

sinusoidal voltage with a frequency equal to an integer multiple of the fundamental frequency of the supply voltage. Harmonic voltages can be evaluated

– individually by their relative amplitude (U_h) related to the fundamental voltage U_1, where h is the order of the harmonic,

– globally, for example by the total harmonic distortion factor THD, calculated using the following expression:

$$THD = \sqrt{\sum_{h=2}^{40} (u_h)^2}$$

NOTE Harmonics of the supply voltage are caused mainly by network users' non-linear loads connected to all voltage levels of the supply network. Harmonic currents flowing through the network impedance give rise to harmonic voltages. Harmonic currents and network impedances and thus the harmonic voltages at the supply terminals vary in time.

3.22
interharmonic voltage

sinusoidal voltage with a frequency between the harmonics, i.e. the frequency is not an integer multiple of the fundamental

NOTE Interharmonic voltages at closely adjacent frequencies can appear at the same time forming a wide band spectrum.

3.23
voltage unbalance

condition in a polyphase system in which the r.m.s. values of the line-to-line voltages (fundamental component), or the phase angles between consecutive line voltages, are not all equal. The degree of the inequality is usually expressed as the ratios of the negative and zero sequence components to the positive sequence component

[IEV 161-08-09 modified]

NOTE 1 In this European Standard, voltage unbalance is considered in relation to three-phase systems and negative phase sequence only.

NOTE 2 Several approximations give reasonably accurate results for the levels of unbalance normally encountered (ratio of negative to positive sequence components), e.g.:

$$\text{voltage unbalance} = \sqrt{\frac{6 \times (U_{12}^2 + U_{23}^2 + U_{31}^2)}{(U_{12} + U_{23} + U_{31})^2} - 2}$$

where U_{12}, U_{23} and U_{31} are the three line-to-line voltages.

3.24
mains signalling voltage

signal superimposed on the supply voltage for the purpose of transmission of information in the public distribution network and to network users' premises. Three types of signals in the public distribution network can be classified:

– **ripple control signals:** superimposed sinusoidal voltage signals in the range of 110 Hz to 3 000 Hz;

– **power-line-carrier signals:** superimposed sinusoidal voltage signals in the range between 3 kHz to 148,5 kHz;

– **mains marking signals:** superimposed short time alterations (transients) at selected points of the voltage waveform.

4 Low-voltage supply characteristics

4.1 Power frequency

The nominal frequency of the supply voltage shall be 50 Hz. Under normal operating conditions the mean value of the fundamental frequency measured over 10 s shall be within a range of:

– for systems with synchronous connection to an interconnected system:
50 Hz ± 1 % (i.e. 49,5 Hz... 50,5 Hz) during 99,5 % of a year;
50 Hz + 4 % / - 6 % (i.e. 47 Hz... 52 Hz) during 100 % of the time;

– for systems with no synchronous connection to an interconnected system (e.g. supply systems on certain islands):
50 Hz ± 2 % (i.e. 49 Hz... 51 Hz) during 95 % of a week;
50 Hz ± 15 % (i.e. 42,5 Hz... 57,5 Hz) during 100 % of the time.

4.2 Magnitude of the supply voltage

The standard nominal voltage U_n for public low voltage is U_n = 230 V, either between phase and neutral, or between phases

– for four-wire three phase systems:
U_n = 230 V between phase and neutral;

– for three-wire three phase systems:
U_n = 230 V between phases.

NOTE In low voltage systems declared and nominal voltage are equal.

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4.3 Supply voltage variations

4.3.1 Requirements

The voltage variation should not exceed ± 10 %.

Situations like those arising from faults or voltage interruptions, the circumstances of which are beyond the reasonable control of the parties, are excluded.

NOTE 1 Overall experience has shown, that sustained voltage deviations of more than ± 10 % over a longer period of time are extremely unlikely, although they could theoretically be within the given statistical limits of 4.3.2. Therefore, in accordance with relevant product and installation standards and application of IEC 60038 end users' appliances are usually designed to tolerate supply voltages of ± 10 % around the nominal system voltage, which is sufficient to cover an overwhelming majority of supply conditions. It is expected to be neither technically nor economically viable to generally give appliances the ability to handle supply voltage tolerances broader than ± 10 %. If, in single cases, evidence is given, that the magnitude of the supply voltage could depart beyond this limit for a longer period of time, additional measures should be taken in cooperation with the local network operator, depending on a risk assessment. The same applies in cases, where specific appliances have an increased sensitivity with respect to voltage variations.

NOTE 2 In cases of electricity supplies in remote areas with long lines or not connected to a large interconnected network, the voltage could be outside the range of U_n + 10 % / - 15 %. Network users should be informed of the conditions.

4.3.2 Test method

Under normal operating conditions,

- during each period of one week 95 % of the 10 min mean r.m.s. values of the supply voltage shall be within the range of U_n ± 10 %, and
- all 10 min mean r.m.s. values of the supply voltage shall be within the range of U_n + 10 % / - 15 %.

4.4 Rapid voltage changes

4.4.1 Single rapid voltage change

A rapid voltage change of the supply voltage is mainly caused either by load changes in network users' installations or by switching in the system.

Under normal operating conditions a rapid voltage change generally does not exceed 5 % U_n but a change of up to 10 % U_n with a short duration might occur some times per day in some circumstances.

NOTE A negative voltage change resulting in a voltage less than 90 % U_n is considered a supply voltage dip (see 4.5).

4.4.2 Flicker severity

Under normal operating conditions, in any period of one week the long term flicker severity caused by voltage fluctuation should be $P_{lt} \leq 1$ for 95 % of the time.

NOTE Reaction to flicker is subjective and can vary depending on the perceived cause of the flicker and the period over which it persists. In some cases $P_{lt} = 1$ gives rise to annoyance, whereas in other cases higher levels of P_{lt} are found without annoyance.

4.5 Supply voltage dips

Voltage dips are generally caused by faults occurring in the network users' installations or in the public distribution network. They are unpredictable, largely random events. The annual frequency varies greatly depending on the type of supply system and on the point of observation. Moreover, the distribution over the year can be very irregular.

Indicative values:
Under normal operating conditions the expected number of voltage dips in a year may be from up to a few tens to up to one thousand. The majority of voltage dips have a duration less than 1 s and a retained voltage greater than 40 %. However, voltage dips with greater depth and duration can occur infrequently. In some areas voltage dips with a retained voltage between 85 % and 90 % of U_n can occur very frequently as a result of the switching of loads in network users' installations.

4.6 Short interruptions of the supply voltage

Indicative values:
Under normal operating conditions the annual occurrence of short interruptions of the supply voltage ranges from up to a few tens to up to several hundreds. The duration of approximately 70 % of the short interruptions may be less than one second.

NOTE In some documents short interruptions are considered as having durations not exceeding one minute. But sometimes control schemes are applied which need operating times of up to three minutes in order to avoid long voltage interruptions.

4.7 Long interruptions of the supply voltage

Accidental interruptions are usually caused by external events or actions which cannot be prevented by the DNO. It is not possible to indicate typical values for the annual frequency and durations of long interruptions. This is due to wide differences in system configurations and structure in various countries and also because of the unpredictable effects of the actions of third parties and of the weather.

Indicative values:
Under normal operating conditions the annual frequency of voltage interruptions longer than three minutes may be less than 10 or up to 50 depending on the area.

Indicative values are not given for **prearranged interruptions**, because they are announced in advance.

4.8 Temporary power frequency overvoltages between live conductors and earth

A temporary power frequency overvoltage generally appears during a fault in the public distribution network or in a network user's installation, and disappears when the fault is cleared. Under these conditions, the overvoltage may reach the value of the phase-to-phase voltage (up to max. 440 V in 230/400 V networks) due to a shift of the neutral point of the three-phase voltage system, the actual value depending upon the degree of load unbalance, and the remaining impedance between the faulty conductor and earth.

The duration is limited by the time taken for the MV protection and circuit breaker to clear the fault, typically no more than 5 s.

Under certain circumstances, a fault occurring upstream of a transformer may produce temporary overvoltages on the LV side for the time during which the fault current flows. Such overvoltages do not generally exceed 1,5 kV r.m.s.

4.9 Transient overvoltages between live conductors and earth

Transient overvoltages at the supply terminals generally do not exceed 6 kV peak.

NOTE 1 The rise time can cover a wide range from milliseconds down to much less than a microsecond. However, for physical reasons transients of longer durations usually have much lower amplitudes. Therefore, the coincidence of high amplitudes and a long rise time is extremely unlikely.

NOTE 2 The energy content of a transient overvoltage varies considerably according to the origin. An induced overvoltage due to lightning generally has a higher amplitude but lower energy content than an overvoltage caused by switching, because of the generally longer duration of such switching overvoltages.

NOTE 3 LV Installations and end users' appliances are designed, in accordance with standard EN 60664-1, to withstand transient overvoltages in an overwhelming majority of situations. Where necessary, according to IEC 60364-4-44, surge protective devices should be selected, according to IEC 60364-5-53, to take account of the actual situations. This is assumed to cover the induced over-voltages due to both lightning and switching.

4.10 Supply voltage unbalance

Under normal operating conditions, during each period of one week, 95 % of the 10 min mean r.m.s. values of the negative phase sequence component (fundamental) of the supply voltage shall be within the range 0 % to 2 % of the positive phase sequence component (fundamental). In some areas with partly single phase or two phase connected network users' installations, unbalances up to about 3 % at three-phase supply terminals occur.

NOTE In this European Standard only values for the negative sequence component are given because this component is the relevant one for the possible interference of appliances connected to the system.